SINAN COUNTY DOMESTIC WASTE LANDFILL BIOGAS POLLUTION CONTROL AND COMPREHENSIVE UTILIZATION PROJECT

Project title	Sinan County domestic waste landfill biogas pollution control and comprehensive utilization project
Project ID	5274
Crediting period	16-January-2024 to 15-January-2034(both dates are included)
Original date of issue	18-September-2024
Most recent date of issue	18-September-2024
Version	01
VCS Standard Version	4.7
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Sinan County domestic waste landfill biogas pollution control and comprehensive utilization project (hereafter referred to as the project) is located at Sinan county landfill site, Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P.R. China. The project is constructed and operated by Sinan County BCCY New Energy Co., Ltd.

Sinan County landfill site started collecting and handling waste at the 2014.

The project started to construct on 10-October-2023. The project has a total designed capacity 1.5 MW (3*0.5 MW) and combusts the LFG, which contains nearly 50% of methane, to generate electricity and export it to the Southern China Power Grid (SCPG). Total electricity supplied to the grid in the crediting period will be 66,251 MWh. VCUs will be claimed from the methane destroyed and the grid electricity displaced. The estimated average GHG emission reductions of the project are 37,961 tCO_2e and the total GHG emission reductions and removals are 379,619 tCO_2e during the fixed 10 years crediting period.

Scenario existing prior to the implementation of the project (the same as baseline scenario):

LFG from Sinan County landfill site is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the SCPG, which is dominated by fossil fuel based power plants.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	16-January-2024 to 15-January-2034 (both dates are included), the crediting period is ten years.	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	Ten years

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	Scope 1: energy industries (renewable-/non-renewable sources) Scope 13: Waste handling and disposal
Project activity type	The project is neither a AFOLU project nor a grouped project

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.1 of VCS Standard (version 4.7), the scope of the VCS Program includes:

VCS Standard Scope	Fulfilled (Yes/No)	Justification
(1) The seven Kyoto Protocol greenhouse gases.	Yes	The project is a LFG recovery to power project. Thus, the project is applicable to this scope.
(2) Ozone-depleting substances (ODS).	N/A	The project activity does not involve any Ozone depleting substances.
(3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	N/A	The project activity is supported by the CDM methodology.
(4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Yes	The applied methodologies AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0) of the project are methodologies approved under CDM Program, which is a VCS approved GHG program.
(5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A	This is not a Jurisdictional REDD+ program nor a nested REDD+ project, hence this eligibility criterion is not applicable.

Furthermore, the project does not belong to the projects excluded in Table 1 of VCS Standard (Version 4.7).

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

Thus, the project is eligible under the scope of VCS program.

According to 3.1 of VCS standard (Version 4.7), establishing a consistent and standardized certification process is critical to ensuring the integrity of VCS projects. Accordingly, certain high-level requirements must be met by all projects, as set out below.

Reference to VCS Standard (Version 4.7)	Requirements	Justification
3.1.1	Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.	The project meets all applicable rules and requirements set out under the VCS Program, including this document.
3.1.2	Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website.	The project applies the methodologies AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0) along with tools or modules as applicable developed under the United Nations Clean Development Mechanism, which can be used for projects and programs registering with VCS as per requirements available on the Verra website.
3.1.3	Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22 below. Projects shall update to the latest version of the methodology when reassessing the baseline or renewing a crediting period.	The project applies the methodologies AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0), which are the latest version.
3.1.4	Projects and the implementation of project	According to the environment impact assessment report

	activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	(EIA) of the project, the project complies with all Chinese relevant laws and regulations.
3.1.5	Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or carbon dioxide removals).	Not applicable as there is no model needs to be chosen by the project proponent as per the applied methodologies AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0).
3.1.6	Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Methodology Requirements.	Not applicable as there are no third-party default factor or standard need to be chosen by the project proponent as per the applied methodologies AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0).
3.1.7	Where the rules and requirements under an approved GHG program conflict with the rules and	Not applicable. The project obeys the rules and requirements of the VCS Program

	requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	
3.1.8	Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	Not applicable as there is no specified capacity limits or any other relevant requirements set out as per the applied methodology and tools.
3.1.9	Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and Effective Dates or equivalent for other program documents, and are listed in a companion Summary of Effective Dates document which corresponds with each update.	The project has followed the latest requirements of Verra.

1.4.2 AFOLU project eligibility

The project is not an AFOLU project. Thus, this section is not applicable.

1.4.3 Transfer project eligibility

The project is not a transfer project. Thus, this section is not applicable.

1.5 Project Design

Indicate if the project has been designed as:

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

The project is not a grouped project. Thus, this section is not applicable.

1.6 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Minister of carbon emission department
Address	10/F, Kailin Building, 99 Ruyi West Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	lwang@bccynewpower.com

1.7 Other Entities Involved in the Project

Organization name	Sinan County BCCY New Energy Co., Ltd.
Role in the project	Project Operation Entity
Contact person	Yaya Xu
Title	Project Manager
Address	10/F, Kailin Building, 99 Ruyi West Road, Zhengzhou, China
Telephone	+86 371 65521780
Email	xuyaya@bccynewpower.com

1.8 Ownership

Sinan County BCCY New Energy Co., Ltd. is the sole project owner of the project. The Project Filing Certificate and approval of Environmental Impact Assessment (EIA) and the Business License of the project owner are the evidences for legislative right. Sinan County BCCY New Energy Co., Ltd. is a wholly-owned subsidiary of Henan BCCY Environmental Energy Co., Ltd., which is responsible for the acquisition of approvals, construction and operation of project. It obeys the management of Henan BCCY Environmental Energy Co., Ltd., including application for VCS carbon emission reduction project. Therefore, the project proponent is Henan BCCY Environmental Energy Co., Ltd.

1.9 Project Start Date

Project start date	16-January-2024
Justification	According to VCS standard, the project start date is the date on which activities that lead to the generation of GHG emission reductions or removals are implemented. The project started operation on 16-January-2024. Thus, this date is the start date of the project.

1.10 Project Crediting Period

Crediting period	☐ *Seven years, twice renewable* ☒ *Ten years, fixed* ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*
Start and end date of first or fixed crediting period	16-January-2024 to 15-January-2034(both dates are included)

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)

16-January-2024 to 31-December-2024	40,682
01-January-2025 to 31-December-2025	44,468
01-January-2026 to 31-December-2026	47,173
01-January-2027 to 31-December-2027	48,832
01-January-2028 to 31-December-2028	43,626
01-January-2029 to 31-December-2029	39,165
01-January-2030 to 31-December-2030	34,735
01-January-2031 to 31-December-2031	30,577
01-January-2032 to 31-December-2032	26,623
01-January-2033 to 31-December -2033	22,935
01-January-2034 to 15-January-2034	803
Total estimated ERRs during the first or fixed crediting period	379,619
Total number of years	10
Average annual ERRs	37,961

1.12 Description of the Project Activity

The GHG emission reductions will be achieved through combustion of the recovered methane gas by gas engines, which would be otherwise emitted to the atmosphere, and the generation of electricity from the landfill gas, which is supplied by the Southern China Power Grid (SCPG) prior to the implementation of the project.

Description of the technology in the project

The project consists in LFG collection, transmission and treatment system, with subsequent electricity generation and grid connection system.

LFG collection system

LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

LFG treatment system

Prior to electricity generation, LFG is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

3 gas engines of 0.5 MW rated power each are fed with the LFG and generate electricity, which is then exported to the grid. The project does not include any flare system. All LFG was used to generate electricity and there was no surplus LFG generated. During the operation period, the LFG flow can be adjusted based on LFG demand of the project. When the engines are scheduled to shut down, the LFG collection system reduces the gas in advance. There was no LFG collected and left in the entire project system during the planned shutdown. At the beginning of the design process, the design took into account the length of the pipeline and the safety features, and the pipeline has the capacity to store the LFG. At the same time, the maintenance personnel will handle the malfunction as soon as possible. So, the LFG won’t pass into the atmosphere.

Table 1.1 Equipment technical parameters[2]

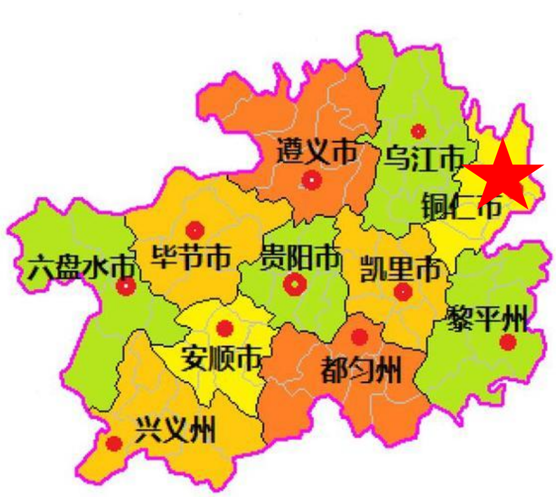
Main Equipment	Parameter	Value
Generator set	Type	500GF-NK
	Manufacturer	CNPC JICHAI POWER EQUIPMENT COMPANY
	Rated capacity	500 kW
	Rated rotation speed	1000 r/min
	Capacity factor	0.8 (lagging)
	Lifetime	15 years

1.13 Project Location

The project is located at Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P. R. China. The geographical coordinates of the project site are longitude 108°11′17.100″ E and latitude 27°55′50.765″ N. Figure 1.1 shows the location of the Project.

[2] The data are from Generator set nameplate and technical agreement.

Map of People’s Republic of China



Map of Guizhou Province

the location of the project site

Figure 1.1 Location of the project

1.14 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity (the same as baseline scenario)

- LFG from Sinan County landfill site is released into the atmosphere.
- Equivalent electricity generated by the project is supplied by the Central China Power Grid, which is dominated by fossil fuel based power plants.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Chinese relevant laws and regulations. Mainly include:

1. Renewable Energy Law of the People's Republic of China;
2. National Action plan for the collection and Utilization of municipal landfill gas;
3. Law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Waste;
4. Catalogue for the Guidance of Industrial Structure Adjustment (2011 version);
5. Hunan Province "14th Five-Year" renewable energy development plan

The project obtained the approval letters from local governmental authorities: Investment Supervisory Department, as well as Environment Protection Agency (EPA). The two approvals well demonstrate that local governments permit the construction of the project. Hence, the project is in compliance with laws, status and other regulatory frameworks.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project is located at Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P. R. China. The project captures the landfill gas and generates electricity, which is fed into the Southern China Power Grid, displacing energy that would otherwise be generated by fossil fuels.

The primary objective of the project is to avoid greenhouse gases emission through landfill gas capture, purification, and injection in a grid connection system, while contributing to the environmental, social, and economic sustainability by minimizing global climate changes and local air pollution. Additional benefits include the increase of employment opportunities through full-time and permanent positions, etc.

The Project will contribute to sustainable development in the following ways:

- SDG 13 Climate Action: The project will recover and destroy landfill gas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030;

- SDG 7 Affordable and Clean Energy: Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Southern China Power Grid and further help to achieve the national action to promote renewable energy development;
- SDG 8 Decent Work and Economic Growth: The project will provide new jobs opportunities and increase tax revenue, which will have a positive effect on the local economy.

Table1.2: Sustainable Development Contributions

SDG Target	SDG Indicator	Project activity contribution
13.0	Tonnes of greenhouse gas emissions avoided or removed	The project avoided anthropogenic emissions of greenhouse gases (GHG) of 379,619 tCO_2e during the crediting period (16-January-2024 to 15-January-2034).
8.5	Total number of job opportunities	The project activity employs 6 permanent and full-time employees through its operational company Sinan County BCCY New Energy Co., Ltd. during the crediting period (16-January-2024 to 15-January-2034).
7.2	MWh of renewable energy generated	Total 66,251 MWh of renewable energy generated during the crediting period (16-January-2024 to 15-January-2034).

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as not AFOLU project.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information available.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The purpose of the project is to utilize LFG to produce electricity, which is supplied to SCPG. The entire project must adhere to local regulations, rules, and obtain necessary permissions. The following steps were adapted to identify stakeholders (based on Section 3.18 of VCS Standard—version 4.7). Step 1: List all individuals or groups who may have an impact on or be affected by the project. Step 2: Conduct a well-being ranking of local or community stakeholders. Step 3: Analyze the interests, motivations for participation, and relationships with other stakeholders of each stakeholder group. Step 4: Analyze the influence and importance of each potential stakeholder group. Step 5: Publish the results of the identified stakeholders. If anyone else believes they are one of the project's stakeholders, they can directly contact the project supporters for consultation. The final stakeholders of this project are as followed: 1. local residents 2. local authorities 3. employees of Sinan County comprehensive administrative law enforcement Bureau (the operator of Sinan County landfill site) 4. employees of Sinan County BCCY New Energy Co., Ltd. 5. the project proponent
Legal or customary tenure/access rights	The project is located at Sinan County landfill site and does not involve the land issues. According to the project cooperation agreement signed between the project proponent and Sinan County comprehensive administrative law enforcement Bureau (the operator of Sinan County landfill site),

	the project proponent has the right to use their LFG for power generation.
Stakeholder diversity and changes over time	Stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time.
Expected changes in well-being	1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; 2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 66,251 MWh, which could replace the equivalent electricity from fossil fuel based grid.
Location of stakeholders	The location of stakeholders is Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P. R. China.
Location of resources	The location of resources is at Sinan County landfill site, Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P. R. China.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	03-November-2023
Stakeholder engagement process	The stakeholders were informed about the stakeholder meeting through posters on 17-October-2023. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and VCS application of the project. The stakeholder consultancy meeting for the parties interested in the project was organized at Sinan County landfill site on 03-November-2023 to collect opinions from all stakeholders, such as representatives of local residents and local authorities, Sinan County comprehensive administrative law enforcement Bureau (the operator of Sinan County landfill site), the

	project proponent and so on. The project proponent appointed one person to make a meeting minute. Bulletin and meeting are in Chinese. The stakeholder meeting was attended by 15 people, five of whom were women.
Consultation outcome	During the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project proponent will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures. In general, all participants have positive perception on project design and implementation. The project would actually facilitate the development of the local economy and increase the income of local residents. During the stakeholder consultancy meeting, the project proponent assured everyone that they would comply with labor laws and ensure the legitimate rights and interests of workers. The project proponent also introduced the VCS validation and verification process to everyone.
Ongoing communication	Expect for introducing the project information, ongoing communication and grievance mechanism were also explained during the local stakeholder consultation meeting. The project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness. Meanwhile, a grievance expression book will be put in the office of the project, which is used to collect all stakeholders' views about the project. Project proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make

	comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible. Until now, no comments or Input / Grievance Expression received from stakeholder.
Stakeholder input	The project did not receive major negative input during the consultation. The project design need not to have any updates. All stakeholders were pleased with the development of the project. The project would actually facilitate the development of the local economy and increase the income of local residents.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Based on the principle of friendly consultation and on the basis of truly and fully expressing their respective wishes, the project proponent and Sinan County comprehensive administrative law enforcement Bureau (the operator of Sinan County landfill site) signed a cooperation agreement on this project. The feasibility study report of the project was approved by the Development and Reform Bureau of Sinan County on 02-August-2023. An environmental impact assessment (EIA) report of the project was approved by Ecological Environment Bureau of Tongren City on 11-October-2023. According to stakeholder consultation meetings, local residents have expressed support for the project. In general, the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders. There are not ongoing or unresolved conflicts during the project validation period.
Outcome of FPIC	The feasibility study report of the project was approved by the Development and Reform Bureau of Sinan County on 02-August-2023.

	An environmental impact assessment (EIA) report of the project was approved by Ecological Environment Bureau of Tongren City on 11-October-2023. The project is located at Sinan County landfill site, Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P. R. China. According to the project cooperation agreement signed between the project proponent and Sinan County comprehensive administrative law enforcement Bureau on 21-July-2023, the land used in this project is Sinan County landfill site, which does not occupy the ecological red line, and does not exceed the range of landfill land red line. Based on questionnaire surveys and stakeholder consultation meeting, local residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	If there are conflicts of interest and dissatisfaction among stakeholders, they can choose to directly appeal to the project proponent or initiate an appeal meeting, which is usually the most effective way to solve the problem. Stakeholders can file appeals or complaints during the meeting through the phone number of the relevant contact person assigned by the project proponent. In addition, the progress of the project is made public on the VCS Pipeline website, accepting public feedback. If stakeholders have appeals and complaints, they can directly provide feedback to Verra. Their opinions will be taken into consideration, and the project proponent will promptly respond and provide feedback to Verra. In terms of government regulation, sometimes complaint mechanisms at the government level can also serve as an effective solution. This project is supervised by the local Ecological Environment Bureau, Development and Reform Commission, and

	Guizhou Resources and Social Security Bureau. The above institutions all have public complaint telephone channels, through which stakeholders can provide feedback on their opinions. The project proponent will respond to it and provide feedback on improvement to government regulatory agencies. In the project, the project proponent has designated a staff member to record and collect dissatisfaction and complaints related to conflicts of interest, who plays an important role in handling these common conflicts and appeals. Once a report regarding the appeal is received, the project proponent will contact and discuss with relevant communities or other stakeholders within 24 hours striving to propose a solution and mediation plan within a week. In any of the above channels, once the project proponent receives a report on the appeal, the project proponent should contact and discuss with the relevant community or other stakeholders within 24 hours, striving to propose a solution and mediation plan within one week. If the problem cannot be resolved within 30 days, it is recommended that stakeholders consider filing a lawsuit.
Grievance redress procedure	The above appeal and compensation procedures were concluded through joint discussions with stakeholders at a stakeholder meeting.

2.1.5 Public Comments

This project is in the public comment period and no comments have been received.

Comments received	Actions taken
/	/

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project proponent has successfully operated several recovery to power plants in China. It has also successfully developed carbon projects related to LFG recovery to power under VCS programme (VCS reference number: 2353, 2731, 2840). Therefore, the PP has expertise and experience in implementing similar project activities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; 2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 66,251 MWh, which could replace the equivalent electricity from fossil fuel based grid. The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. Thus, there are no natural and human-induced risks to stakeholders' wellbeing.
Risks to stakeholder participation	No risk identified	Based on the stakeholder meeting, the stakeholders involved in the project were very supportive of the project. The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China.

Working conditions	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. The Project has DCS system. Therefore, most of the employees work in indoor environment (at the office), instead of having to stand outside. In case of on-site monitoring and device maintenance, the project proponent provides safety protection equipment.
Safety of women and girls	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published "Women's Rights Protection Law"[3] to provide support and protection to women and girls.
Safety of minority and marginalized groups, including children	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published protection laws on the minors[4]. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not

[3] https://www.gov.cn/guoqing/2021-10/29/content_5647634.htm

[4] http://www.gov.cn/xinwen/2020-10/18/content_5552113.htm

		employ and is not complicit in any form of child labour. According to the Employee list, it could be confirmed that the Project does not employ any child labour.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	An environmental impact assessment (EIA) report of the project was approved by Ecological Environment Bureau of Tongren City on 11-October-2023. During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction would be treated according to the measures in EIA report, and there would be no significant impact on the environment. During the operation period, all the mitigation measures proposed by EIA report would be implemented and the following key aspects would be addressed: **Waste water** The wastewater generated by the project mainly includes condensate wastewater generated by the collection system and pretreatment system, domestic sewage and circulating cooling water drainage of the cooling treatment system. The domestic sewage is collected in the septic tank after pretreatment, and the surrounding villagers are regularly entrusted to clear it for agricultural fertilizer use. The condensate waste water is produced in a very

		small amount (1 m^3/a), which is collected by natural evaporation in the condensate collection tank and then sent to the landfill for landfill treatment. The circulating cooling water of the generator set evaporates naturally during use, and the circulating water is not discharged. **Waste Gas** The waste gas is mainly generated by the burning landfill gas of the generator set, and the main waste gas pollutants are smoke, SO_2 and NO_x. The concentration of waste gas after treatment meets the limit requirements of Table 2 of the Emission Standard for Air Pollutants in Thermal Power Plants (GB-13223-2011). **Noise pollution** The noise is mainly generated by generator sets, draught fans, air compressors and other equipment, through the selection of low-noise equipment, the installation of equipment and equipment connection using shock pad or flexible joint measures, the noise contribution value of the factory boundary can meet the requirements of class 2 of Industrial Enterprise factory boundary Environmental noise Emission Standard (GB12348-2008). **Solid waste**

		Solid waste includes domestic waste, general solid waste and hazardous waste. Among them, the filter waste belongs to the general solid waste. Waste oil, waste lubricating oil drums and waste oily rags are hazardous wastes. Domestic waste and filter waste are transported directly to the landfill site for treatment. Waste oil, waste lubricating oil drums and waste oily rags are temporarily stored in the hazardous waste room after collection, and then regularly entrusted with qualified units for disposal.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[5]	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	No discrimination was occured. China has published “Labor Law”[6] and “Women’s Rights Protection Law”, which could prohibit such phenomena. The project abides the rules of equality and does not involve and is not complicit in any form of discrimination.
Sexual harassment	No risk identified	No sexual harassment was occured. China has published “Labor Law” and “Women’s Rights Protection Law”, which could prohibit such phenomena. The

[5] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

[6] https://www.gov.cn/banshi/2005-05/25/content_905.htm

		program adheres to the principle of respecting women, does not involve and is not complicit in any form of sexual harassment.
Equal pay for equal work	No risk identified	As a project combined of positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the LFG sources to replace fossil fuel power. Qualified local residents, both men and women, are recruited to work for the project. The project follows the provisions of the "Constitution"[7] and "Labor Law" on equal pay for men and women for equal work. During stakeholders' consultation process, comments were collected from the local people, including both men and women.
Gender equity in labor and work	No risk identified	As a project combined of positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the LFG sources to replace fossil fuel power. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in "Women's Rights Protection Law" to provide support and benefits to women.

[7] https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

		Qualified local residents, both men and women, are recruited to work for the project. The project follows the provisions of the “Constitution”[8] and “Labor Law” on equal pay for men and women for equal work. During stakeholders’ consultation process, comments were collected from the local people, including both men and women. The project abides the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in “Labor Law” and “Women’s Rights Protection Law”.
Forced labor	No risk identified	There will be no forced labor in this project. China has promulgated the “Labor Law” to protect the legitimate rights and interests of workers and prohibit forced labor.
Child labor	No risk identified	China has published protection laws on the minors. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labour. According to the Employee list, it could be confirmed that the Project does not employ any child labour.
Human trafficking	No risk identified	The project will not use victims of human trafficking. Human trafficking is a serious crime that is strictly regulated and combated in the Chinese legal system. The

[8] https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

		relevant laws and regulations mainly include the “Criminal Law”[9]and the “Criminal Procedure Law”[10].

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Human rights protection is formally embodied in the regulations in China. China fully recognizes international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties--the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. This project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women. The project proponent and its activities are strictly regulated by local and national regulations, which obligate the project activities

[9] https://flk.npc.gov.cn/detail2.html?ZmY4MDgxODE3OTZhNjM2YTAxNzk4MjJhMTk2NDBjOTI

[10] https://flk.npc.gov.cn/detail2.html?ZmY4MDgwODE2ZjEzNWY0NjAxNmYxZDFiODFiMDEzNTE%3D

	to in line with the international human right law and conventions. The legal approval by authorities demonstrate that the project does not and will not violate the regulations. According to the relevant provisions of the Labor Law and Women's Rights Protection Law of China, the project continues to recognize, respect, and promote the protection of the rights of IPs, LCs, and customary rights holders.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	According to the EIA report, no indigenous peoples present in or within the area of influence of the Project. According to the EIA report, the Project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The feasibility study report of the project was approved by the Development and Reform Bureau of Sinan County on 02-August-2023. An environmental impact assessment (EIA) report of the project was approved by Ecological Environment Bureau of Tongren City on 11-October-2023. The project is located at Sinan County landfill site, Dingjia Village, Nixi Village, Dahebei Town, Sinan County, Tongren City, Guizhou Province, P. R. China. According to the project cooperation

	agreement signed between the project proponent and Sinan County comprehensive administrative law enforcement Bureau on 21-July-2023, the land used in this project is Sinan County landfill site, which does not occupy the ecological red line, and does not exceed the range of landfill land red line. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable as no property rights involved for the project.
Summary of the benefit sharing plan	Not applicable as no property rights involved for the project.
Approval and dissemination of benefit sharing plan	Not applicable as no property rights involved for the project.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project is LFG recovery and utilization project. According to the EIA report, it does not affect biodiversity and ecosystems.

Soil degradation and soil erosion	No risk identified	The Project's purpose is to supply electricity from LFG. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.
Water consumption and stress	No risk identified	The Project's purpose is to supply electricity from LFG. Therefore, the impact of the Project on water consumption and stress is negligible.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Species and habitat	Not applicable as the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.
Areas needed for habitat connectivity	Not applicable as the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	According to EIA, the project isn't adjacent to habitats for rare, threatened, or endangered species.
Areas for habitat connectivity	No risk identified	According to EIA, the project doesn't belong to areas for habitat connectivity.

2.4.2 Introduction of Species

According to EIA, no planting or species introduction are in the project. This section is left as N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation

N/A	N/A	N/A	N/A

According to EIA, no invasive species exist in the project area. Therefore, the following table isn't applicable.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	N/A

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risk identified	According to EIA, no invasive species exist in the project area.

2.4.3 Ecosystem Conversion

The project isn't ARR, ALM, WRC or ACoGS project. Hence, not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.G.	Landfill methane recovery[11]	10.0

[11] https://cdm.unfccc.int/methodologies/DB/0KHNES8D09H134V3TZDQ47C3LQL3H2

Methodology	AMS-I.D.	Grid connected renewable electricity generation[12]	18.0
Tool	04	Emissions from solid waste disposal sites[13]	8.1
Tool	07	Tool to calculate the emission factor for an electricity system[14]	07.0
Tool	32	Positive lists of technologies[15]	4.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AMS-III.G.	2.This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	Applicable. The project consists in capturing landfill gas (which contains methane) from the Sinan county landfill site, which is used for disposal of residues from human activities.
	3.Different options to utilise the recovered landfill gas as detailed in paragraph 4 for “AMS-III.H. : Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this	Applicable. The project utilizes the recovered LFG to generate electrical energy directly, i.e. 4(a) of AMS-III.H. (version 19.0).

[12] https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK

[13] http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v6.0.1.pdf/history_view

[14] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v2.2.1.pdf/history_view

[15] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-32-v1.pdf/history_view

	methodology. The relevant procedures in AMS-III.H. shall be followed in this regard. Paragraph 4 of AMS-III.H.: The recovery biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users;	

	or(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users (d) Hydrogen production; Use as fuel in transportation applications after upgrading.	
	4.Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.	Applicable. The project results in aggregate emission reduction of less than 60kt CO_2 equivalent annually from all Type III components.
	5.The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity	Applicable. The implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in the Sinan county landfill site.
	6.This methodology is not applicable if the management of the solid waste disposal site SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include,	Not applicable. The management of the SWDS in the project activity will not be deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity.

	for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS or changing the shape of the SWDS to increase methane production	
AMS-I.D.	4.This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or;Involve a replacement of (an) existing plant(s).	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project, which corresponds to point (a).
	5.Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is	Not applicable. The project is not a hydro power plant.

	increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than $4W/m^2$; The project activity results in new reservoirs and the power density of the powerplant, as per definitions given in the project emissions section, is greater than $4W/m^2$.	
	6.If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel. the capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable. The project does not use non-renewable components nor co-fires fossil fuels. Anyway, the total installed capacity is below 15MW.
	7.Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable. The project only involves electricity generation.
	8.In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.

	should be physically distinct from the existing units.	
	9.In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable. The project does not involve retrofit, rehabilitation, or replacement.
	10.In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-I.C. Thermal energy production with or without electricity shall be explored.	Applicable. The project installed two engines to combusted LFG of Sinan county landfill site, which mainly contains 50% methane.
Tool 04	(a) Application A: The GS project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001:	Applicable. The project collects the methane from the Sinan county landfill site and uses it to generate the electricity.

	Flaring or use of landfill gas). The methane is generated from waste disposed in the past, including prior to the start of the GS project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (GS PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e. g. measuring the amount of methane captured from the SWDS).	
	Application B: The GS project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	Not applicable. The project activity mitigates methane emissions from Sinan county Landfill by capturing and utilizing the methane for power generation. This tool is only applied for ex ante estimation of the emissions in the PD.

Tool 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. This tool is applied to estimate the OM, BM and/or CM.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity	Applicable. The emission factor for the project electricity system is calculated for grid power plants only.

	system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
	In case of VCS projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable. The project electricity system is located in China.
	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	Not applicable. The project does not involve biofuels.
Tool 32	The use of this methodological tool is not mandatory for the project participants of a GS project activity or GS PoA for demonstrating their additionality.	Applicable. The project collects the LFG from the Sinan county landfill site and uses it to generate the electricity, with a total capacity of 1.5 MW, which conforms to 5.1.1 (a) “The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW”.
	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. This tool is applied in conjunction with small-scale methodology AMS-III.G. (version 10.0).

3.3 Project Boundary

According to the adopted methodology, the project boundary is determined as follows:

- The project boundary is the site of the project activity where the gas is captured and destroyed/used.

- If the electricity for project activity is sourced from grid or electricity generated by the LFG captured would have been generated by power generation sources connected to the grid, the project boundary shall include all the power generation sources connected to the grid to which the project activity is connected.
- If the electricity for project activity is from a captive generation source or electricity generated by the captured LFG would have been generated by a captive power plant, the captive power plant shall be included in the project boundary.

The project boundary is the site of the project activity, Sinan County landfill site, where the LFG is captured and used, and also includes all the power sources connected to the SCPG, which expands throughout Guangdong Province, Guangxi Zhuang Autonomous Region, Yunnan Province, Guizhou Province and Hainan Province.

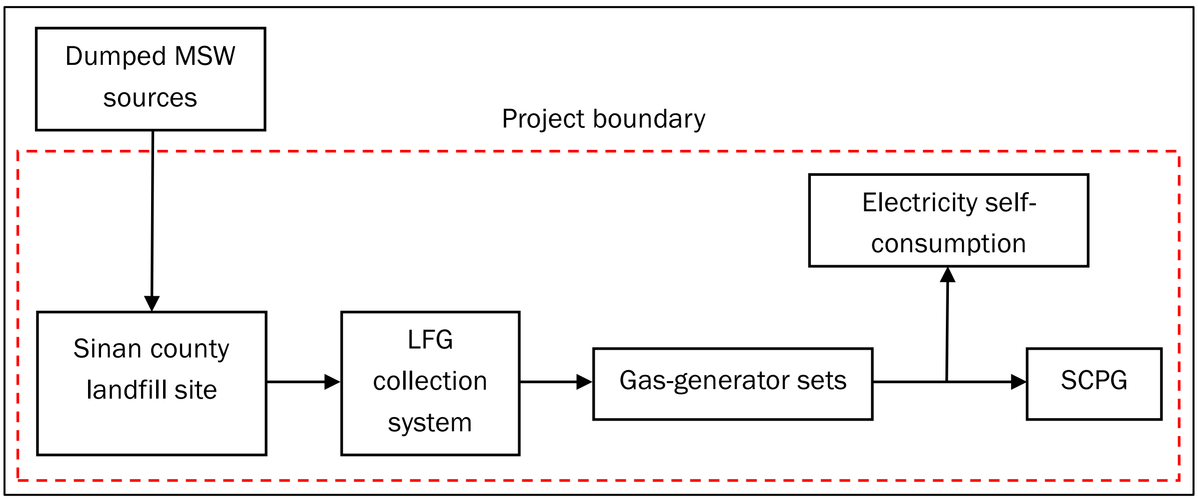


Figure 3.1 the Diagram of Project Boundary

Table 3.1 Summary of greenhouse gases and sources included in and excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO_2	No	CO_2 emissions from decomposition of organic waste are not accounted since the CO_2 is also released under the project activity
		CH_4	Yes	The major source of emissions in the baseline
		N_2O	No	N_2O emissions are small compared to CH_4 emissions from SWDS. This is conservative

Source		Gas	Included?	Justification/Explanation
	Emissions from electricity generation	CO_2	Yes	Major emission source
		CH_4	No	Excluded for simplification. This is conservative
		N_2O	No	Excluded for simplification. This is conservative
	Emissions from heat generation	CO_2	No	No heat generation is included in the project activity
		CH_4	No	No heat generation is included in the project activity
		N_2O	No	No heat generation is included in the project activity
	Emissions from the use of natural gas	CO_2	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
		CH_4	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
		N_2O	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO_2	No	Excluded for no fossil fuel is used by the project activity
		CH_4	No	Excluded for no fossil fuel is used by the project activity
		N_2O	No	Excluded for no fossil fuel is used by the project activity

Source		Gas	Included?	Justification/Explanation
	Emissions from electricity consumption due to the project activity	CO_2	Yes	May be an important emission source
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from flaring	CO_2	No	Excluded for simplification
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO_2	No	No distribution of LFG using trucks and dedicated pipelines
		CH_4	No	No distribution of LFG using trucks and dedicated pipelines
		N_2O	No	No distribution of LFG using trucks and dedicated pipelines

3.4 Baseline Scenario

Landfill gas: In the absence of the project, MSW of Sinan County landfill site is left to decay within the project boundary, and methane is emitted to the atmosphere directly without any recovery and utilization.

Electricity: The project is a new grid-connected renewable power unit.

According to the section 5.3 of the methodology AMS-III.G. and 5.2 of methodology AMS-I.D., the baseline scenario is:

LFG from Sinan County landfill site is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the SCPG, which is dominated by fossil fuel based power plants.

3.5 Additionality

The project uses the following steps to demonstrate additionality. The project meets the additionality criteria as outlined in the VCS methodology as:

Step 1: Regulatory Surplus: the project is not mandated by any legal requirements. Please refer to section 3.5.1 below for details.

Step 2: Positive List: the project conforms to the positive list as stipulated in the applies the simplified procedures in Section 5.2 of the methodology AMS-III.G. (version 10.0) and CDM "TOOL32: Positive lists of technologies" (version 04.0). Please refer to section 3.5.2 below for details.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

According to the relevant applicable laws and regulations at the time of submission of registration, i.e. Renewable Energy Law of the People's Republic of China which came into effect on 01-January-2006, Agenda for China's 14th Five-Year Plan (2021-2025), there are no new national and/or sectoral policies that could affect the baseline scenario during the crediting period. Therefore, the project is surplus to regulatory requirements.

3.5.2 Additionality Methods

According to the section 5.2 of the methodology AMS-III.G., the project chooses simplified procedures to demonstrate additionality. Tool32 (Positive lists of technologies) is used to demonstrate additionality.

Requirements of tool 32 (Positive lists of technologies)	The project

5.1Waste handling and disposal 5.1.1Landfill gas recovery and its gainful use The project activities and PoAs at new or existing landfills (greenfield or brownfield)are deemed automatically additional if it is demonstrated that prior to the implementation of the project activities and PoAs the landfill gas(LFG)was only vented and or flared (in the case of brownfield projects)or would have been only vented and/or flared(in the case of greenfield projects)but not utilized for energy generation, and that under the project activities and PoAs any of the following conditions are met: (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption (c)The LFG is flared	The LFG from Sinan county landfill site is vented to the atmosphere prior the implementation of the project for safety concerns. Total nameplate capacity of the project is 1.5 MW, which is below 10MW.

Conclusion: the project is automatically additional. The project is not enforced by law. There is a cap & trade scheme in China. However, the project activity is not included in the mandatory emission control scheme since the scheme only covers the high-emission industries, such as power generation with fossil fuels. The project is in compliance with applicable China's legal requirements and obtains the legal approvals. So the project is additional.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project utilizes the LFG for generation to substitute the equivalent electricity supplied by the grid, resulting in CH_4 and CO_2 emissions, which will be calculated as follows according to the methodology AMS-III.G. and AMS-I.D.

Baseline emissions associated with the SWDS:

The project utilizes the LFG for generation to substitute the equivalent electricity supplied by the grid, resulting in CH_4 and CO_2 emissions, which will be calculated as follows according to the methodology AMS-III.G. and AMS-I.D.

$$BE_{y,1} = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4} \tag{1}$$

Where:

$BE_{y,1}$	Baseline emissions associated with the SWDS in year y (tCO_2e/yr)
η_{PJ}	Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used
$BE_{CH4,SWDS,y}$	Methane emission potential of a solid waste disposal site (in tCO_2e)
OX	Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering waste) (dimensionless). A default value of 0.1 may be used
$F_{CH4,BL,y}$	Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (t CH_4). The relevant procedures in “ACM0001: Flaring or use of landfill gas” may be followed, as well as taking into account the compliance with the relevant local laws and regulation if such laws and regulations exist
GWP_{CH4}	Global Warming Potential for methane

Ex ante determination of $BE_{CH4,SWDS,y}$

According to the AMS-III.G., $BE_{CH4,SWDS,y}$ is calculated using the methodological tool "Emission from solid waste disposal sites", which is calculated as follows:

$$BE_{CH4,SWDS,y} = \varphi_y \times (1 - f_{y)} \times GWP_{CH4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^{y} \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{k_j}) \quad (2)$$

Where:

$BE_{CH4,SWDS,y}$	=	Baseline methane emissions occurring in the year y
φ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emission of methane to the atmosphere in year y
GWP_{CH4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of the methane in the SWDS gas (volume fraction)
$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
MCF_y	=	Methane correction factor for year y
$W_{j,x}$	=	Amount of organic waste type j disposed/prevented from disposal in the SWDS in the year y (t)
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction for year y
k	=	Decay rate for the waste type(1/yr)
j	=	Type of residual waste or types of waste in the MSW
x	=	Years of the crediting period for which waste is disposed at the SWDS, extending from the first year in the time period (x=1) to year y (x=y)

y	=	Years of the crediting period for which methane emissions are calculated

Determination of $F_{CH4,BL,y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as requirement in this section). The four cases in Table B.2 are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Table B.2 Cases for determining methane captured and destroyed in the baseline

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Currently China has regulations in place to deal with the management of landfills and to encourage utilization of LFG. Those regulations are:

“Standard for Pollution Control on the Landfill Site of Municipal Solid Waste” (GB 16889-2008), which became effective in 2008, issued by the Environment Protection Administration.

“Technical Code for Municipal Solid Waste Sanitary Landfill” (GB 50869-2013), issued by the Ministry of Construction in 2013.

According to item 5.15 of GB16889-2008, if the designed landfill capacity is more than 2.5 million tons and the landfill thickness is more than 20m, methane utilization facilities or flare burning facilities shall be built to treat the landfill gas containing methane. For municipal solid waste landfills smaller than the above scale, technologies that can effectively reduce methane generation and emission shall be adopted or flare combustion facilities shall be used to treat methane containing landfill gas.

Item 11.1.1 of GB 50869-2013 stipulates that the landfill site must be equipped with effective landfill gas drainage facilities to prevent the natural accumulation and migration of landfill gas, causing fire and explosion. Item 11.1.3 stipulates that if the landfill does not have the conditions for landfill gas utilization, the flare method shall be adopted for combustion treatment, and the process that can effectively reduce the generation and emission of methane shall be adopted.

The old landfills that are not safe and stable should be equipped with effective landfill gas drainage facilities. Among them, item 11.1.1 is mandatory and must be strictly implemented.

In fact, the LFG of Wenxi landfill sites emitted to atmosphere without LFG capture system prior the implementation of the project. Therefore, Case 2 listed in the Table B.2 is applicable for the project.

The requirements above don't specify any amount or percentage of LFG that should be destroyed. In this situation:

$$F_{CH4,BL,y} = F_{CH4,BL,R,y} \quad (3)$$

$$F_{CH4,BL,R,y} = 0.2 \times F_{CH4,PJ,capt,y} \quad (4)$$

Where:

$F_{CH4,BL,R,y}$	=	Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (t CH_4/yr)
$F_{CH4,PJ,capt,y}$	=	Amount of methane in the LFG which is captured in the project activity in year y (t CH_4/yr)

The annual waste tipping has been estimated in the FSR, which is listed as follow:

Table 4.2 Estimated amounts of waste that have been dumped into the landfill

Year	Annual landfilled MSW— Wx (tons/a)
2014	36,500
2015	54,385
2016	72,270
2017	82,855
2018	86,505
2019	87,600
2020	87,965
2021	86,870
2022	83,950
2023	87,600
2024	87,965

2025	88,330
2026	89,060
2027	89,425

According to the FSR, the contents of the MSW are listed below:

Table 4.3 the waste contents of the landfills

Component	**Weight Fraction (wet waste)**
W_1-Wood and wood products	0.1800
W_2-Pulp, paper and cardboard	0.1500
W_3-Food, food waste, beverages and tobacco	0.4236
W_4-Textiles	0.0109
W_5-Garden, yard and park waste	0.0306
W_6-Glass, plastic, metal other inert	0.2049

The values for DOC_j and k_j are shown in table 4.4.

Table 4.4 DOC_j and k_j values for type of waste j

Component	**DOC_j**	**k_j**
W_1-Wood and wood products	0.43	0.03
W_2-Pulp, paper and cardboard	0.40	0.06
W_3-Food, food waste, beverages and tobacco	0.15	0.185
W_4-Textiles	0.24	0.06
W_5-Garden, yard and park waste	0.20	0.01
W_6-Glass, plastic, metal other inert	0.00	0

The parameters adopted to calculate the methane generation and collection are listed below:

Table 4.5 The values of parameters

Parameter	Value
ϕ	0.75
GWP_{CH4}	25
	28
f	0
MCF	1.0
DOC_f	0.5
F	0.5
OX	0.1

Table 4.6 Ex-ante estimation of $BE_{y,1}$

Year	$BE_{CH4,SWDS,y}$ (Methane emission)	$F_{CH4,PJ,y}$ (CH_4 destroy)	$F_{CH4,BL,y}$ (20% discount)	$BE_{CH4,y}$ (ER from CH_4)
Unit	tCO_2e	tCH_4	tCH_4	tCO_2e
16-January-2024 to 31-December-2024	53,080	1,611	361	36,996
01-January-2025 to 31-December-2025	58,019	1,761	395	40,439
01-January-2026 to 31-December-2026	61,549	1,868	419	42,899
01-January-2027 to 31-December-2027	64,338	1,953	391	44,843
01-January-2028 to 31-December-2028	57,479	1,745	349	40,062
01-January-2029 to 31-December-2029	51,602	1,567	313	35,966
01-January-2030 to 31-December-2030	45,765	1,389	278	31,898

01-January-2031 to 31-December-2031	40,286	1,223	245	28,079
01-January-2032 to 31-December-2032	35,076	1,065	213	24,448
01-January-2033 to 31-December -2033	30,218	917	183	21,062
01-January-2024 to 15-January-2024	1,060	32	6	738
Total	**498,473**	**15,132**	**3,152**	**347,430**
Average	**49,847**	**1,513**	**315**	**34,743**

Baseline emissions associated with electricity generation ($BE_{EC,y}$)

$$BE_{y,2} = EG_{PJ,y} \times EF_{grid,y} \quad (5)$$

Where:

$BE_{y,2}$	=	Baseline emissions associated with electricity generation in year y (t CO_2)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh)
$EF_{grid,y}$	=	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad (6)$$

Where:

$EG_{PJ,facility,y}$	=	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y(MWh)

Calculation of $EF_{grid,y}$

According to the AMS-III.G., $EF_{grid,y}$ is calculated using the "Tool to calculate the emission factor for an electricity system"($EF_{grid,y}$=$EF_{grid,CM,y}$).

Project participants shall apply the following six steps:

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the selected method;

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identity the relevant electricity systems

National Center for Climate Change Strategy and International Cooperation has published a delineation of the project electricity system and connected electricity systems, so the project adopts the delineation of project electricity system and connected electricity system published by National Center for Climate Change Strategy and International Cooperation. The power generated by the project displaces the equivalent electricity generated by the Southern China Power Grid. The Southern China Power Grid is a larger regional grid, which consists of five sub-grids: Guangdong Province, Guangxi Zhuang Autonomous Region, Yunnan Province, Guizhou Province and Hainan Province.

In addition, there is net imported power to the Southern China Power Grid from the Northeast China Power Grid and Central China Power Grid. According to the "Tool to calculate the emission factor for an electricity system", use one of the following options to determine the CO_2 emission factor for net electricity imports from a connected electricity system:

(a) 0tCO_2/MWh; or

(b) The simple operating margin emission rate of the exporting grid, determined as described in Step 4 section 6.4.1, if the conditions for this method, as described in Step 3 below, apply to the exporting grid; or

(c) The simple adjusted operating margin emission rate of the exporting grid, determined as described in Step 4 section 6.4.2 below; or

(d) The weighted average operating margin (OM) emission rate of the exporting grid, determined as described in Step 4 section 6.4.4 below.

The PD will choose option (b) to calculate the CO_2 emission factor for net electricity imports from the North China Power Grid.

According to the available data of 2018-2020, the corresponding marginal emission factors of electricity are calculated, and weighted average is carried out to obtain the marginal emission factors of electricity.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional).

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Following the calculation of the National Center for Climate Change Strategy and International Cooperation, and statistical data is available, Option I is chosen.

Step 3. Select a method to determine the operating margin (OM)

"Tool to calculate the emission factor for an electricity system (Version 7.0)" offers four methods for the calculation of the operating margin emission factor(s) ($EF_{grid,OM,y}$):

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

Method (a) -Simple OM is chosen for calculation and low-cost/must-run resources constitute less than 50% of the total grid generation in average of the five most recent years[16].

For simple OM, the emission factor can be calculated using either of the two following data vintages:

(a) Ex ante option: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the GS-PDD to the DOE for

[16] 2021 Bulletin on the Baseline Emission Factors of the China Grids

validation. For off-grid power plants, use a single calendar year within the 5 most recent calendar years prior to the time of submission of the VCS-PDD for validation;

(b) Ex post option: If the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year y-1 may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all crediting periods

Project participant employs ex ante option for its operation margin calculation.

Step 4. Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units. The 2021 OM emission factors of North China Power Grid, Northeast Power Grid, East China Power Grid and Northwest Power Grid use the simple OM method.

The simple OM may be calculated by one of the following two options:

(a) Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; or

(b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:

(i) The necessary data for Option A is not available: and

(ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(iii) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

Since the data of each power plant/unit is unavailable, Option A is not applicable to the project. The project adopts Option B to calculate the operating margin emission factor ($EF_{grid,OM,y}$) of SCPG.

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y} \quad (7)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year y (tCO_2/MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fossil fuel type i in year y (tCO_2e/GJ), 2006 IPCC Guidelines for default values
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must run power plants/units in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

For Central China Power Grid and Southern Power Grid, use the adjusted simple OM method to calculate their OM emission factor. The adjusted simple OM divides the generator sets into two categories: low cost/must run power plants and all the other power plants except low cost/ must run power plants. According to the weighted average of each unit's annual power supply emission factor, the probability weights of the two units participating in generation scheduling (λ and 1-λ) are obtained. Finally, the unit power supply emission factor of the two types of generating units is adjusted according to the participation in generation. The probability weights of degrees are weighted and averaged to obtain the adjusted simple OM. The specific calculation formula is as follows:

$$EF_{grid,OM-adj,y} = (1-\lambda_y) \times EF_{grid,OMsimple,y} + \lambda_y \times \frac{\sum_k (EG_{k,y} \times EF_{EL,k,y})}{\sum_k EG_{k,y}} \quad (8)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year y (tCO_2/MWh), according to the equation (7)

$EG_{k,y}$	=	Net electricity generated and delivered to the grid by low-cost/must run power plants/units type k in year y (MWh)
$FE_{EL,k,y}$	=	CO_2 emission factor by low-cost/must run power plants/units type k in year y (tCO_2/MWh)
k	=	Type of low-cost/must run power plants/units in year y
λ_y	=	Probability weight of low-cost/must run power plants/units participating in generation scheduling in year y
y	=	The relevant year as per the data vintage chosen in Step 3

In the above formula, the unit power supply emission factor of the transferred electricity in the low-cost/must run unit adopts the transferred electricity OM emission factor of power grid. Hydropower, wind power, photovoltaic, geothermal power, low-cost biomass power, and nuclear power, the emission factor per unit power supply is 0 tCO_2/MWh. For low cost/must run units involved in power generation, the probability weight λ for scheduling is obtained by consulting "Tool to calculate the emission factor for an electricity system (Version 7.0)" Appendix 2 of Table 1.

Because there is also electricity exchange between Central China Power Grid and Southern Power Grid, the adjusted simple OM factor of the two power grids is obtained by solving a system of linear equations.

Regarding parameter selection, local values of $NCV_{i,y}$ and $EF_{CO2,i,y}$ should be used where available. If no such values are available, IPCC default values are preferable. The Net Calorific Value ($NCV_{i,y}$) of each type of fossil fuel used in the calculation comes from China Electric Yearbook (2020-2022), Electricity Industry Statistics (2019-2021) and China Energy Statistic Yearbook (2020-2022). Emission factor ($EF_{CO2,i,y}$) of each type of fossil fuel come from IPCC 2006 default values.

The operating margin emission factor for 2019, 2020 and 2021 are calculated based on the data above. The three-year average is calculated as a weighted average of the emission factors. The Operational Margin Emission Factor is 0.7738tCO_2/MWh.

Step 5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1: For the crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of GS PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

(b) Option 2: For the crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The project applies option 1 to calculate the build margin emission factor ex-ante.

According to the “Tool to calculate the emission factor for an electricity system”, the sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently ($SET_{5\ units}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

(b) Determine the annual electricity generation of the project electricity system, excluding power units registered as GS project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as GS project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET\text{-}\geq 20\%}$, in MWh);

(c) From $SET_{5\text{-}units}$ and $SE_{T\geq 20\ percent}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use $SE_{Tsample}$ to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as GS project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set (SET_{sample}) the annual electricity generation ($AEG_{SET\text{-}sample}$, in MWh);

If the annual electricity generation of that set is comprised at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET\text{-}sample} \geq 0.2 \times AEG_{total}$), then use the sample group SET_{sample} to calculate the build margin. Ignore steps (e) and (f).

(e) Include in the sample group SET_{sample} the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set (SET_{sample}>10yrs)

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \qquad (9)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (t CO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

Under the current circumstances in China, the power plants consider the Build Margin data as important business data and will not have them published. Therefore, it is difficult to obtain the data of five power plants that have been put into operation most recently or the newly installed power plant capacity additions in the electricity system that comprise 20% of the system generation.

According to the instructions of China DNA, for the determination of the set of samples, a sample merging processing in some degree has been adopted due to that the power generation data, energy consumption data or thermal efficiency data of each plant cannot be consulted in the public statistical data. In this calculation, the newly-installed power units in the past years are classified by year, province and power generation technology, and the same type of newly installed power units in the same province and in the same year are bundled as a "newly-installed power units".

The power generation of each "newly-installed power units" in the most recent year y is estimated based on its installed capacity and the number of power generation utilization hours in year y. The formula is as follows:

$$EG_{m,y} = CAP_m \times H_{m,y} \tag{10}$$

Where:

$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW)
$H_{m,y}$	=	The number of power utilization hours (h) of electricity generated and delivered to the grid by power unit m in year y. And it selects the average utilization hours of similar units in the province in which it is located in year y;
y	=	Most recent year for which data is available
m	=	The sample group of power units

The power unit m is selected from the "newly-installed power plants" in the most recent year y (For the calculation of the grid BM in 2021, the y is equal to 2019) to the "newly-installed power plants" in the earlier year, until the cumulative power generation reaches 20% of the total power generation in the year y (y=2019).

Since the newly-installed power units of the same type (k) in the same province (A) and the same year (t) are bundled into the "newly-installed power units", the CAP_m is equal to the statistical data of recent installed capacity of a given unit type(k) in a given year(t) in a given province (A).

$$CAP_m = CAP_m|_{m=(A,t,k)} = CAP_{A,t,k} \tag{11}$$

Where:

CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW), and m is equivalent to an established combination of (A, t, k)
$CAP_{A,t,k}$	=	Capacity of newly-installed power units of a given province (A), given year (t), and given unit type (k) (MW)
A	=	It is the various provincial regions covered by the regional power grid
t	=	It is the sampling year of the "newly-installed power units". For the calculation of the grid BM in 2021, t is equal to 2019, 2018, until the units that comprise at least 20 percent of the system generation in 2019.
k	=	It is the power generation technology classification of "newly-installed power units", which is divided into: hydro-power, coal-thermal power, gas-thermal power, oil-thermal power, Waste-thermal power plant, other thermal power[17], nuclear power, wind power, solar power, and others[18].

As per tool, the CO_2 emission factor of each power unit m ($EF_{EL,m,y}$) should be determined as per the tool in Step 4 section 6.4.1 for the simple OM, using Options A1, A2 or A3, using for y the most recent historical year for which electricity generation data is available, and using form the power units included in the build margin.

Because current statistics data cannot separate each power plant, for a power unit m, only data on electricity generation and the fuel types used is available. So, the option A2 is selected, the emission factor should be determined based on the CO_2 emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \tag{12}$$

Where:

[17] refers to waste heat and pressure, straw, bagasse, forest wood power generation

[18] refers to power generation such as geothermal energy and ocean energy.

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type i used in power unit m in year y (tCO_2/GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit m in year y (dimensionless)
3.6		Conversion coefficient of thermal work equivalent of electricity (GJ/MWh)

According to Equation (10), the unit electricity emission factor of the hydro-power, nuclear power, wind power, solar power, other thermal power, and others power generation technology in the "newly-installed power units" samples are is zero. The emission factor per unit of electricity for power generation from coal, gas, oil and electricity waste power is calculated based on Equation (10). Since the average net energy conversion efficiency of each sample ($\eta_{m,y}$) cannot be obtained, the power supply thermal efficiency of the best commercialized technology of coal, gas, oil and waste power ($\eta_{Best,m,y}$) is better than $\eta_{m,y}$. It is conservative to use $\eta_{Best,m,y}$ for the calculation of $EF_{EL,m,y}$. Therefore, the formula is as follows:

$$EF_{Best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{Best,m,y}} \qquad (13)$$

Where:

$EF_{Best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year y (tCO_2/MWh)
$\eta_{Best,m,y}$	=	Power generation efficiency of the best technology commercially available in year y (dimensionless)
m	=	Sample of "newly-installed power units" generated by coal, gas, oil and waste incineration in a province in a certain year

The emission factors of coal, natural gas, fuel oil and municipal solid waste (MSW) are sourced from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, and the lower values of the 95% confidence interval are applied to be conservative.

No fuel oil power units were constructed in 2019. The power generation efficiency of the best technology commercially available for fuel oil units comes from Baseline Emission Factors for Regional Grids in China of previous years, i.e., 52.9%.

According to the above steps, the build margin emission factor, $EF_{grid,BM,y}$ of the SCPG is calculated to be 0.1981 tCO_2/MWh, which is confirmed by 2023 Baseline Emission Factors for Regional Power Grids in China published by National Center for Climate Change Strategy and International Cooperation[19].

Step 6. Calculate the combined margin emission factor

Combined Margin emission factor ($EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin emission factor($EF_{grid,OM,y}$) and the build margin emission factor ($EF_{grid,BM,y}$),where the weights ω_{OM} and ω_{BM} ,by default, are 0.5 and 0.5 in the crediting period, and $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ are calculated as described above and are expressed in tCO_2/MWh.

$$EF_{grid,CM,y} = \omega_{OM} \times EF_{grid,OM,y} + \omega_{BM} \times EF_{grid,BM,y} \quad (14)$$

$EF_{grid,CM,y}$=0.5×0.7738+0.5×0.1981=0.48595 (tCO_2/MWh)

The $EF_{OM,y}$, $EF_{BM,y}$ and $EF_{CM,y}$ are ex-ante calculation and are fixed during the crediting period.

The CO_2 emissions from electricity generation by grid-connected power plants under baseline scenario that is displaced by LFG electricity generation under the project activity are tabulated below table 4.6.

Table 4.6 Estimation of baseline emission $BE_{y,2}$

Year	Power Generation	Power suppied to Grid	ER for power replace
Unit	MWh	MWh	tCO_2e
16-January-2024 to 31-December-2024	8,070	7,586	3,686
01-January-2025 to 31-December-2025	8,821	8,292	4,029
01-January-2026 to 31-December-2026	9,358	8,796	4,274
01-January-2027 to 31-December-2027	8,734	8,210	3,989
01-January-2028 to 31-December-2028	7,803	7,335	3,564
01-January-2029 to 31-December-2029	7,005	6,585	3,199

[19]https://mp.weixin.qq.com/s/fJ2SvbdrA7QVnCemI4DFEQ

01-January-2030 to 31-December-2030	6,213	5,840	2,837
01-January-2031 to 31-December-2031	5,469	5,141	2,498
01-January-2032 to 31-December-2032	4,762	4,476	2,175
01-January-2033 to 31-December -2033	4,102	3,856	1,873
01-January-2034 to 15-January-2034	144	135	65
Total	**70,480**	**66,251**	**32,189**
Average	**7,048**	**6,625**	**3,218**

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$BE_y = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4} + EG_{PJ,y} \times EF_{grid,y} \quad (15)$$

The baseline emission reduction is estimated in table 4.7.

Table 4.7 Estimation of baseline emission BE_y

Year	**ER from CH_4**	**ER from power**	**Baseline Emission**
Unit	**tCO_2e**	**tCO_2e**	**tCO_2e**
16-January-2024 to 31-December-2024	36,996	3,686	40,682
01-January-2025 to 31-December-2025	40,439	4,029	44,468
01-January-2026 to 31-December-2026	42,899	4,274	47,173
01-January-2027 to 31-December-2027	44,843	3,989	48,832
01-January-2028 to 31-December-2028	40,062	3,564	43,626
01-January-2029 to 31-December-2029	35,966	3,199	39,165

01-January-2030 to 31-December-2030	31,898	2,837	34,735
01-January-2031 to 31-December-2031	28,079	2,498	30,577
01-January-2032 to 31-December-2032	24,448	2,175	26,623
01-January-2033 to 31-December-2033	21,062	1,873	22,935
01-January-2034 to 15-January-2034	738	65	803
Total	**347,430**	**32,189**	**379,619**
Average	**34,743**	**3,219**	**37,961**

4.2 Project Emissions

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y} \tag{16}$$

Where:

PE_y	=	Project emission in year y (t CO_2e)
$PE_{power,y}$	=	Emission from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO_2e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the landfill gas stream in the year y (t CO_2e)
$PE_{process,y}$	=	Emission from the landfill gas upgrading process in the year y (t CO_2e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

According to AMS-III.G., project emissions from electricity consumption are determined as per the procedures described in AMS-I.D. “Grid connected renewable electricity generation”. According to AMS-I.D., the project is neither involved geothermal power plants nor hydro power plants, project emissions from electricity consumption are identified as 0. Besides, as fossil fuel is not be used by the project, project emissions from fossil fuel consumption are also 0. Therefore, $PE_{power,y}$ is equal to 0.

There is no flare used to destroy the LFG, therefore, $PE_{flare,y}$ is equal to 0.

The project captures LFG for power generation and not involved upgrading process, therefore, $PE_{process,y}$ is equal to 0.

The project emission reduction is estimated as follow:

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y}$$

=0+0+0=0

4.3 Leakage Emissions

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.G. and AMS-I.D.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

According to AMS-III.G., the emission reduction achieved by the project activity can be estimated ex-ante in the PDD by:

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad (17)$$

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$ER_{y,estimated} = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} \times GWP_{CH4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad (18)$$

The ex-ante estimate of the emission reduction is:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
16-January-2024 to 31-December-2024	40,682	0	0	40,682	0	40,682
01-January-2025 to 31-December-2025	44,468	0	0	44,468	0	44,468
01-January-2026 to 31-December-2026	47,173	0	0	47,173	0	47,173
01-January-2027 to 31-December-2027	48,832	0	0	48,832	0	48,832

01-January-2028 to 31-December-2028	43,626	0	0	43,626	0	43,626
01-January-2029 to 31-December-2029	39,165	0	0	39,165	0	39,165
01-January-2030 to 31-December-2030	34,735	0	0	34,735	0	34,735
01-January-2031 to 31-December-2031	30,577	0	0	30,577	0	30,577
01-January-2032 to 31-December-2032	26,623	0	0	26,623	0	26,623
01-January-2033 to 31-December-2033	22,935	0	0	22,935	0	22,935
01-January-2034 to 15-January-2034	803	0	0	803	0	803
Total	379,619	0	0	379,619	0	379,619

Emission reductions (ex-post)

According to AMS-III.G., the actual emission reduction achieved by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calcaulated} = (1 - OX) \times \left(F_{CH4,PJ,y} - F_{CH4,BL,y}\right) \times GWP_{CH4} - PE_y - LE_y \quad (19)$$

Where:

The project utilizes the recovered methane for power generation. Therefore $F_{CH4,PJ,y}$ is calculated as follows, based on the amount of monitored electricity generation without monitoring methane flow and concentration:

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \quad (20)$$

Where:

EG_y	=	Electricity generation in year y (MWh)

3600	=	Conversion factor (1 MWh=3600 MJ)
NCV_{CH4}	=	NCV of methane (MJ/NM^3) use default value: 35.9 MJ/NM^3
EE_y	=	Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel if the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
- Default efficiency of 40 per cent

Therefore:

$$ER_{y,calcaulated} = (1 - OX) \times \left(\frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} - F_{CH4,BL,y}\right) \times GWP_{CH4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad (21)$$

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories[20]
Value applied	0.1
Justification of choice of data or description of	-

[20] https://www.ipcc-nggip.iges.or.jp/public/2006gl/chinese/index.html

measurement methods and procedures applied	
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP_{CH4}
Data unit	$tCO_2e/t\ CH_4$
Description	Global warming potential of CH_4
Source of data	IPCC Fifth Assessment Report (AR5)
Value applied	Default value of 28. Shall be updated according to any future COP/CMP decisions
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\rho_{CH4}(D_{CH4,y})$
Data unit	Tonnes $/m^3$
Description	Density of methane gas at Normal Conditions
Source of data	Tool 08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)
Value applied	0.0007156
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	At the normal conditions (273.15 K and 101,325 Pa)

Data / Parameter	η_{PJ}
Data unit	-
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	Project Feasibility Study Report
Value applied	85%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	φ_y
Data unit	-
Description	The model correction factor to account for model uncertainties
Source of data	Default value of the tool 04 “Emissions from solid waste disposal sites” (version 08.1)
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	Application A is used to decide the value. And the value of dry conditions for used for the project since the MAP/PET>1. The meteorological data for the project site (Sinan County) are: Mean annual temperature (MAT): 17.3°C[21] Mean annual precipitation (MAP): 1,128.6 mm[22] Potential evapotranspiration (PET): 935.3 mm[23]
Purpose of Data	Calculation of baseline emissions
Comments	-

[21] https://www.sinan.gov.cn/zjsn/zrdl/

[22] https://www.sinan.gov.cn/zjsn/zrdl/

[23] https://www.sinan.gov.cn/zwgk/zdlygk/yjgl/yjgg/201709/t20170922_64819689.html

Data / Parameter	f_y
Data unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	The tool 04 "Emissions from solid waste disposal sites" (version 08.1)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	It has been considered in the section 4.1, equation (1). Therefore, f_y=0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$DOC_{f,y}$
Data unit	Weight fraction

Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposed in the SWDS
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCF_y
Data unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	The project using application A to decide the value of MCF_y. The Sinan county landfill site has controlled placement of waste, mechanical compacting and levelling of the waste, So MCF_y value of 1.0 for anaerobic managed solid waste disposal sites should be applied in line with" Emissions from solid waste disposal sites" (version 8.1)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	W_x
Data unit	t
Description	Amount of solid waste type j disposed or prevent from disposal in the SWDS in year y
Source of data	Records from landfill operator
Value applied	Year / Annual landfilled MSW (tons)

	2014	36,500
	2015	54,385
	2016	72,270
	2017	82,855
	2018	86,505
	2019	87,600
	2020	87,965
	2021	86,870
	2022	83,950
	2023	87,600
	2024	87,965
	2025	88,330
	2026	89,060
	2027	89,425

Waste type j	Weight Fraction (% wet waste)
W_1-Wood and wood products	0.1800
W_2-Pulp, paper and cardboard	0.1500
W_3-Food, food waste, beverages and tobacco	0.4236
W_4-Textiles	0.0109
W_5-Garden, yard and park waste	0.0306
W_6-Glass, plastic, metal other inert	0.2049

Justification of choice of data or description of measurement methods and procedures applied	The data FSR of the project.
Purpose of Data	Calculation of baseline emissions

Comments	-

Data / Parameter	DOC_j
Data unit	-
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)
Source of data	The tool 04 "Emissions from solid waste disposal sites" (version 08.1)
Value applied	<table><tr><td>Waste type j</td><td>DOC_j (% wet waste)</td></tr><tr><td>Wood and wood products</td><td>43</td></tr><tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr><tr><td>Textiles</td><td>24</td></tr><tr><td>Garden, yard and park waste</td><td>20</td></tr><tr><td>Glass, plastic, metal, other inert waste</td><td>0</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	Data available are based on the wet condition of the waste.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	k_j
Data unit	1/yr
Description	Decay rate for the waste type j
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)
Value applied	<table><tr><td>Waste type j</td><td>k_j (MAT≤20°C,</td></tr></table>

		MAP/PET>1)
Slowly degrading	Pulp, paper, cardboard (other than sludge)	0.06
	Wood, wood products and straw	0.03
Moderately degrading	Other(non-food) organic putrescible garden and park waste	0.1
Rapidly degrading	Food, food waste, sewage sludge, beverages and tabacco	0.185

Justification of choice of data or description of measurement methods and procedures applied	The meteorological data for the project site (Sinan County) are: Mean annual temperature (MAT): 17.3°C Mean annual precipitation (MAP): 1128.6 mm Potential evapotranspiration (PET): 935.3 mm
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh
Description	Operation margin emission factor of SCPG
Source of data	2023 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.9350
Justification of choice of data or description of measurement methods and procedures applied	The value is published by National Center for Climate Change Strategy and International Cooperation, based on China Electric Yearbook (2020-2022), Electricity Industry Statistics (2019-2021), China Energy Statistic Yearbook (2020-2022) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$

Data unit	tCO_2/MWh
Description	Build margin emission factor of SCPG
Source of data	2023 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.3020
Justification of choice of data or description of measurement methods and procedures applied	The value is published by National Center for Climate Change Strategy and International Cooperation, based on China Electric Yearbook (2020-2022), Electricity Industry Statistics (2019-2021), China Energy Statistic Yearbook (2020-2022) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$F_{CH4,BL,y}$
Data unit	tCH_4/yr
Description	Amount of methane in the LFG which is flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Annually
Value applied	The data below are calculated ex ante Year: 16-January-2024 to 31-December-2024 Amount of methane in the LFG which is flared due to a requirement (tCH_4): 361

01-January-2025 to 31-December-2025	395
01-January-2026 to 31-December-2026	419
01-January-2027 to 31-December-2027	391
01-January-2028 to 31-December-2028	349
01-January-2029 to 31-December-2029	313
01-January-2030 to 31-December-2030	278
01-January-2031 to 31-December-2031	245
01-January-2032 to 31-December-2032	213
01-January-2033 to 31-December -2033	183
01-January-2034 to 15-January-2034	6

Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001"Flaring or use of landfill gas" (version 19.0)

Data / Parameter	$\rho_{reg,y}$
Data unit	Dimensionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.
Description of measurement methods and procedures to be applied	-

Frequency of monitoring/recording	Annually
Value applied	20%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001"Flaring or use of landfill gas" (version 19.0)

Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh
Description	Quantity of net electricity generated supplied by the project plant/unit in year y
Source of data	Electricity meter E2
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter (bi-directional) installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.
Value applied	The data below are calculated ex ante <table><tr><td>Year</td><td>Quantity of net electricity generated supplied by the project plant (MWh)</td></tr><tr><td>16-January-2024 to 31-December-2024</td><td>7,586</td></tr><tr><td>01-January-2025 to 31-December-2025</td><td>8,292</td></tr><tr><td>01-January-2026 to 31-December-2026</td><td>8,796</td></tr></table>

	01-January-2027 to 31-December-2027	8,210
	01-January-2028 to 31-December-2028	7,335
	01-January-2029 to 31-December-2029	6,585
	01-January-2030 to 31-December-2030	5,840
	01-January-2031 to 31-December-2031	5,141
	01-January-2032 to 31-December-2032	4,476
	01-January-2033 to 31-December -2033	3,856
	01-January-2034 to 15-January-2034	135
Monitoring equipment	Electricity meter E2 shown in the monitoring system as below.	
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.	
Purpose of data	Calculation of baseline emissions	
Calculation method	-	
Comments	This parameter represents the net electricity supplied to a grid, which is the difference of the electricity exported (EG_{out}) and imported (EG_{in}) by Data will be aggregated monthly. Therefore, the net electricity supplied to the grid was calculated as: $EG_{out} - EG_{in}$.	

Data / Parameter	EG_y
Data unit	MWh
Description	Electricity generation in year y
Source of data	Electricity meter E1
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter installed at the project site. All data will be monitored and archived electronically.

Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.
Value applied	The data below are calculated ex ante <table><tr><td>Year</td><td>Electricity generated in year y (MWh)</td></tr><tr><td>16-January-2024 to 31-December-2024</td><td>8,070</td></tr><tr><td>01-January-2025 to 31-December-2025</td><td>8,821</td></tr><tr><td>01-January-2026 to 31-December-2026</td><td>9,358</td></tr><tr><td>01-January-2027 to 31-December-2027</td><td>8,734</td></tr><tr><td>01-January-2028 to 31-December-2028</td><td>7,803</td></tr><tr><td>01-January-2029 to 31-December-2029</td><td>7,005</td></tr><tr><td>01-January-2030 to 31-December-2030</td><td>6,213</td></tr><tr><td>01-January-2031 to 31-December-2031</td><td>5,469</td></tr><tr><td>01-January-2032 to 31-December-2032</td><td>4,762</td></tr><tr><td>01-January-2033 to 31-December -2033</td><td>4,102</td></tr><tr><td>01-January-2034 to 15-January-2034</td><td>144</td></tr></table>
Monitoring equipment	Electricity meter E1 shown in the monitoring system as below.
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$D_{CH4,y}$
Data unit	t/m^3
Description	Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m^3). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions
Source of data	Tool 08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)
Description of measurement methods and procedures to be applied	The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed. According to the AMS-III.G., the density of methane is also determined at normal conditions
Frequency of monitoring/recording	-
Value applied	0.0007156
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the landfill gas
Source of data	-
Description of measurement methods and procedures to be applied	The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed. According to the AMS-III.G., there is no need for separate monitoring of pressure and temperature of the landfill gas.
Frequency of monitoring/recording	-

Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	℃
Description	Temperature of the landfill gas
Source of data	-
Description of measurement methods and procedures to be applied	The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed. According to the AMS-III.G., there is no need for separate monitoring of pressure and temperature of the landfill gas.
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EE_y
Data unit	%

Description	Energy Conversion Efficiency of the project equipment
Source of data	-
Description of measurement methods and procedures to be applied	As per paragraph 21 of AMS.III.G. (Version 10.0) Specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for biogas. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
Frequency of monitoring/recording	-
Value applied	40%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

5.3 Monitoring Plan

1. The requirement of monitoring plan

   The project participants will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodology AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0) to ensure project ERs are measurable and real. The monitoring methodology is based on direct measurement of the amount of landfill gas captured and destroyed by the project and electricity generating units.

2. Responsibilities of operational and management structure

   The project participant will implement this monitoring plan. The plan could be revised according to suggestions from DOE and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

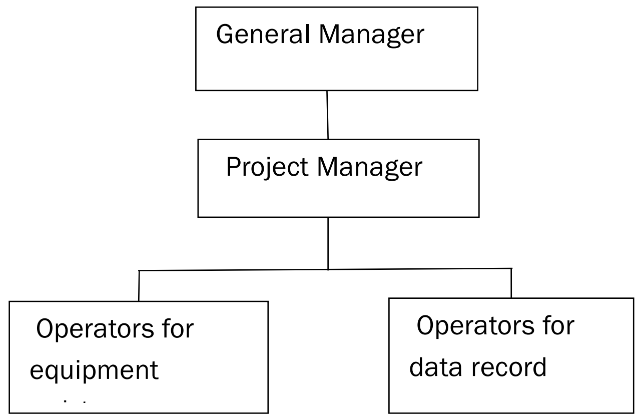


Figure 6.2 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. The general manager ensures that staff in the monitoring system has the ability to deal with the assigned tasks.

(2) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The VCS Manager reports monthly to the General Manager (GM) about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the VCS manager will inform the GM about the situation and work out relevant measures to be taken. The VCS manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system

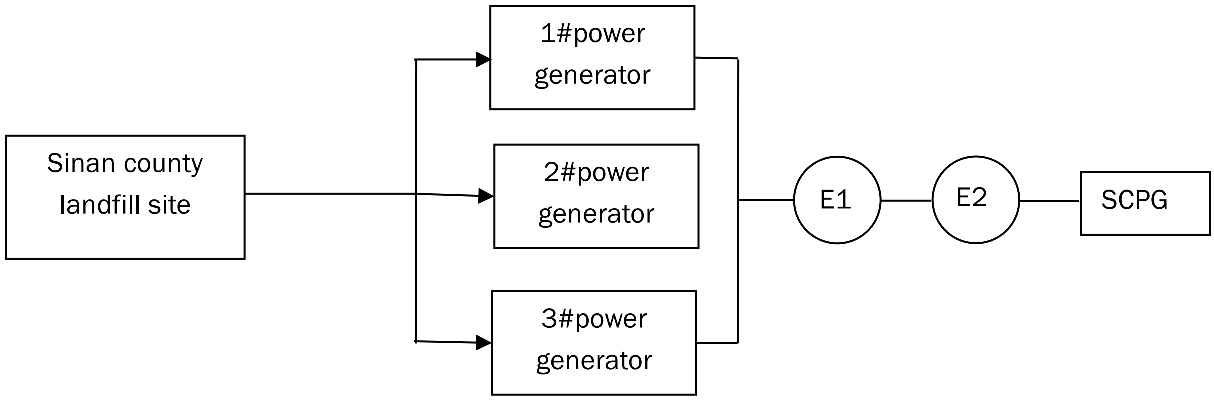


Figure 6.3 Monitoring system

Main parameters:

E1	Electricity meter (accuracy:0.5S) to continuously measure the total electricity by all the generators
E2	Electricity meter (main meter, bi-directional, accuracy:0.5S) to continuously measure the electricity supplied to the SCPG and the electricity supplied by the grid

4.QA/QC

The operator who is responsible for equipment maintenance invites a qualified third party to calibrate the meters (flow meters, gas analyzer and electricity meters). The calibration reports were checked by the project manager and then archived in the project owner archives by the operator. When the data was not available from the main monitoring devices, the data measured by the back-up devices or estimation based on the historical data was used.

The project performance and monitoring data were reported monthly to the General Manager for review and internal audit of calculation and archived and kept for emission reduction calculation yearly during the crediting period and two years after.

5.Emergency procedures

The mistakes occurred in daily operation include problems of the equipment and data. If problems were found and/or monitored, each department reports to the Operation Department, who asked the corrections by providing some suggestions.

As for the data acquisition system, the DCS consists of monitoring and recording the data. Once the system had some problems, the Technical Department and/or factory technicians solved as soon as possible. The DCS data during this period was not adopted.